

Stronger local overlap certificates and distinct zeros of the Riemann zeta function

Felipe Santibanez-Leal 
CAOS Research program

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Abstract

We derive a lower asymptotic proportion $\frac{30945470743359}{36955122080000} = 0.8373797460706156\dots$ of distinct nontrivial zeta zeros, relative to zeros counted with multiplicity, by applying Knausgård's mixed-multiplicity Gram argument to Lavery's seven-point local theorem with unequal gap pressures. The external local theorem is an explicit dependency; its reported Lean and nanoda checks were not rebuilt locally. A separate complete eight-gap interval computation proves a stronger nine-point local inequality at pressure $1/2500$ and gives the independently replayed bound $\frac{3997934614153}{4775507750000} = 0.8371747724947154\dots$. The block transfer uses an elementary spectral dichotomy, without a sharp trace envelope. Independent rational window enclosures, complete closed-cell input comparisons and extracted-source checks support the computations. The analytic input is the corrected integrated unconditional pair-correlation theorem of Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh. The results are asymptotic; they do not prove the Riemann hypothesis or give an effective height.

1 Counts, context and results

Let $N(T)$ count nontrivial zeros ρ of $\zeta(s)$ with $0 < \text{Im } \rho \leq T$, including multiplicity, and let $N_d(T)$ count the distinct points in the same region. Distinctness refers to the whole critical strip. A multiple zero contributes one to N_d ; two zeros symmetric about the critical line contribute two. A bound for N_d/N therefore differs from a bound for simple zeros on the critical line.

The unconditional pair-correlation method supplied the recent simple-zero advances of Alpöge and Furman and Lamzouri [6, 7]. Knausgård [3] retained the Gram defect for critical-line zeros of multiplicity one or two and obtained

$$\liminf_{T \rightarrow \infty} \frac{N_d(T)}{N(T)} \geq \frac{16260119298029}{19426831050000} = 0.83699288145242\dots \quad (1)$$

His matrix inequality, threshold lemma, elementary block dichotomy and counting assembly are the underlying method here. We give the transfer proof with a vector of gap pressures and apply two stronger local inputs.

Theorem 1 (Derived consequence, D). *Using Lavery's universal seven-point local theorem [4] and the corrected integrated BGSTB pair-correlation theorem [1, 2],*

$$\liminf_{T \rightarrow \infty} \frac{N_d(T)}{N(T)} \geq q_7 := \frac{30945470743359}{36955122080000} = 0.8373797460706156\dots \quad (2)$$

A separate machine-verified local certificate, with the same analytic and mixed-Gram inputs, gives

$$\liminf_{T \rightarrow \infty} \frac{N_d(T)}{N(T)} \geq q_9 := \frac{3997934614153}{4775507750000} = 0.8371747724947154 \dots \quad (3)$$

The gain in (2) over (1) is approximately 0.000386865, or 0.0386865 percentage points. The nine-point gain is approximately 0.000181891. These are numerical improvements within an attributed method. Quadratic convex lower bounds, pressure summation and the scalar window identity below are classical. The source comparison is limited to the primary material available on 3 October 2026 and does not establish worldwide priority. The nine-point weights and window come from the source candidate of tawanerguo-cn [5]; the stronger target is proved by the complete computation described in Section 4.

2 A transfer for nonnegative vector pressures

Let f be bounded, even, nonnegative and supported on $[-1/2, 1/2]$, with $\int f = 1$. Write

$$K(x) = \int_{-1/2}^{1/2} f(t) e^{2\pi i x t} dt, \quad w(x) = K(x)^2 \quad (x \in \mathbb{R}),$$

$$H_f = 2 - \left(\int f^2 + \iint |s - t| f(s) f(t) ds dt \right).$$

The kernels used below have positive smooth profiles inside this interval. Assume nonnegative rational weights a_{ij} , $0 \leq i < j \leq r$, satisfy

$$\sum_{i=0}^{r-s} a_{i,i+s} \leq 2 \quad (1 \leq s \leq r). \quad (4)$$

Suppose nonnegative pressures $b_0, \dots, b_{r-1}$ and $\delta > 0$ give the universal local inequality

$$\sum_{i < j} a_{ij} w(g_i + \dots + g_{j-1}) + \sum_{i=0}^{r-1} b_i g_i \geq \delta \quad (g_0, \dots, g_{r-1} \geq 0). \quad (5)$$

Set $B = \sum b_i$. Equal pressures are not required.

Theorem 2 (Mixed-Gram transfer, D). *Let (4)–(5) hold, and let $m > r$ be an integer. Choose $\tau > 0$ and c such that*

$$\begin{aligned} D := \delta(m-r) &\leq \tau^2, & c &\geq \max(1 + \tau, 2 + \tau/2), \\ a := D/m &< 2, & 6c - 7 - c^2 - a &\geq 0, & 4c - 2 - c^2 - 2a &\geq 0. \end{aligned} \quad (6)$$

Under the corrected integrated BGSTB input, with $\beta = B(m-r)/m$, one has

$$\liminf_{T \rightarrow \infty} \frac{N_d(T)}{N(T)} \geq \frac{1 + H_f - \beta}{2 - a}. \quad (7)$$

2.1 Block defect and pressure counting

For $\lambda \geq 0$ put

$$\Psi_\tau(\lambda) = (\lambda - 1)^2 - (\lambda - 1 - \tau)_+^2.$$

It is nonnegative and convex, and $\Psi_\tau(1) = 0$. A Gram matrix U for m ordered points has unit diagonal. Summing (5) over its $m - r$ consecutive windows and using (4) gives

$$\text{tr}(U - I)^2 + P_{\text{block}} \geq D, \quad (8)$$

where P_{block} is the sum of the window pressure charges. If all eigenvalues of U are at most $1 + \tau$, then its clipped defect $\text{tr } \Psi_\tau(U)$ equals $\text{tr}(U - I)^2$. Otherwise one eigenvalue contributes more than τ^2 , while all other terms and pressures are nonnegative. Thus $D \leq \tau^2$ implies

$$\text{tr } \Psi_\tau(U) + P_{\text{block}} \geq D. \quad (9)$$

This is Knausgård's elementary dichotomy [3]. A sharp trace-zero envelope is unnecessary at the parameter choices below.

Order l points $y_0 < \dots < y_{l-1}$ and partition them into complete consecutive m -point blocks for each of the m offsets, leaving shorter endpoint blocks. Convex trace pinching bounds the full Gram defect from below by the sum of the complete-block defects. Omitted points may be pinched into singletons, each contributing zero.

Each possible block start occurs once over all offsets, so the total number of complete blocks is exactly $\max(l - m + 1, 0)$. A fixed r -gap window lies in at most $m - r$ complete blocks. A fixed gap appears in local windows with total pressure at most $\sum_i b_i = B$: each pressure position occurs at most once. Therefore, with span defined as zero for $l \leq 1$, averaging (9) yields

$$\text{tr } \Psi_\tau(U_{\text{full}}) \geq al - \beta(y_{l-1} - y_0) - D. \quad (10)$$

For $l < m$, the right side is nonpositive, so the same assertion follows from nonnegativity. This argument proves the pressure coefficient for all nonnegative vectors. Replacing B by $r \min b_i$ would undercount unequal charges and is not justified.

2.2 The matrix inequalities and multiplicities

For $c > 0$ define $\phi_c(x) = x^2 - (x - c)_+^2$. We recall and prove the two attributed matrix inequalities needed for the count [3].

Lemma 3 (Threshold inequality). *If $P \succeq 0$, Q is Hermitian and has at most b positive eigenvalues, and $\text{tr}(P + Q) = N$, then*

$$\text{tr}(P + Q)^2 \geq 2cN - 2c \text{tr } P + \text{tr } \phi_c(P) - c^2b.$$

Proof. Write $Q = Q_+ - Q_-$ with $Q_+Q_- = 0$. The cross term $\text{tr}((P - Q_-)Q_+) = \text{tr}(PQ_+) \geq 0$. Completing squares on Q_+ gives $\text{tr } Q_+^2 \geq 2c \text{tr } Q_+ - c^2b$. Substituting $\text{tr } Q_+ = N - \text{tr } P + \text{tr } Q_-$ leaves

$$2cN - 2c \text{tr } P + \text{tr } P^2 + \text{tr } Q_-^2 - 2 \text{tr}((P - cI)Q_-) - c^2b.$$

The last quadratic, minimized over positive semidefinite Q_- , is $-\text{tr}((P - cI)_+^2)$. This proves the assertion. $\square$

Lemma 4 (Mixed-multiplicity Gram inequality). *Let U be a Hermitian Gram matrix with unit diagonal, and $D_0 = \text{diag}(d_i)$, $1 \leq d_i \leq 2$. If $c \geq \max(1 + \tau, 2 + \tau/2)$, then*

$$\text{tr } \phi_c(D_0^{1/2}UD_0^{1/2}) - \text{tr } D_0^2 \geq \text{tr } \Psi_\tau(U).$$

Proof. For any Hermitian G, A with $A \preceq 2cI$, diagonalizing G gives

$$\text{tr}(AG) - \frac{1}{4} \text{tr } A^2 \leq \sum_i (A_{ii}\lambda_i - A_{ii}^2/4) \leq \sum_i \phi_c(\lambda_i) = \text{tr } \phi_c(G).$$

The scalar inequality follows by completing a square with $A_{ii} \leq 2c$. Set $X = U - I$, $C = \min(X, \tau I)$ by spectral calculus, $M = D_0^{-1/2}CD_0^{-1/2}$ and $A = 2D_0 + 2M$. Since $C \preceq \tau I$, $A \preceq 2(D_0 + \tau D_0^{-1}) \preceq 2cI$; the convex function $d + \tau/d$ on $[1, 2]$ is bounded by its endpoint maximum. Substitution with $G = D_0^{1/2}UD_0^{1/2}$ and its unit diagonal yields

$$\text{tr } \phi_c(G) - \text{tr } D_0^2 \geq 2 \text{tr}(CX) - \text{tr } M^2 \geq 2 \text{tr}(CX) - \text{tr } C^2 = \text{tr } \Psi_\tau(U).$$

The middle inequality is entrywise: $\text{tr } M^2 = \sum_{ij} |C_{ij}|^2 / (d_i d_j) \leq \text{tr } C^2$. Neither C nor M needs to be positive semidefinite. $\square$

Let h count critical-line points with multiplicity at least three and k count functional-equation pairs off the line. The remaining $l = N_d - h - 2k$ points have multiplicity one or two. For the smoothed complex-zero operator constructed below, their contribution P is positive semidefinite and the remainder Q has at most $h + k$ positive eigenvalues. The nonzero spectrum of P is that of $D_0^{1/2}UD_0^{1/2}$. If n_1, n_2 count the low points of multiplicity one and two, then $\text{tr } D_0^2 = n_1 + 4n_2 = 3 \text{tr } P - 2l$.

Combining Lemmas 3 and 4, the remaining mass is at least $3h + 2k$. Because $c \geq 2$, increasing its multiplicities only strengthens the lower estimate. One obtains

$$\text{tr}(P + Q)^2 \geq 3N - 2N_d + (6c - 7 - c^2)h + (4c - 2 - c^2)k + \text{tr } \Psi_\tau(U). \quad (11)$$

For example a high point of multiplicity $j \geq 3$ contributes $(2c - 3)j + 2 - c^2 \geq 6c - 7 - c^2$; an off-line pair of multiplicity $j \geq 1$ contributes $(4c - 6)j + 4 - c^2 \geq 4c - 2 - c^2$.

2.3 The analytic input and order of limits

Choose even smooth positive cutoff profiles $f_\epsilon = \eta_\epsilon^2$ supported strictly inside $(-1/2, 1/2)$, normalized to integral one, and converging to f in L^1 and L^2 , with uniformly bounded profiles. Put $L = \log T$ and $z_\rho = i(\rho - 1/2)L/(2\pi)$ for zeros with $0 < \text{Im } \rho \leq T$. Let $F_z(u) = \eta_\epsilon(u)e^{2\pi izu}$, using the inner product linear in its second slot. The operator

$$\mathcal{A}_\epsilon = \sum_z \nu_z F_z \langle F_{\bar{z}}, \cdot \rangle$$

is Hermitian by the functional equation. Its trace is N , and

$$E_\epsilon := \text{tr } \mathcal{A}_\epsilon^2 = \sum_{\rho, \rho'} \nu_\rho \nu_{\rho'} K_\epsilon(z_\rho - z_{\rho'})^2.$$

All operators act on a finite-dimensional span. An off-line pair contributes $\nu(vw^* + wv^*) = \nu((v + w)(v + w)^* - (v - w)(v - w)^*)/2$, which has at most one positive eigenvalue. Together with the high critical-line points this justifies the positive-rank bound used in (11).

BGSTB's integrated Lemma 5 [1] sums over complex zeros with weight $\omega(\rho - \rho') = 4/(4 - (\rho - \rho')^2)$. Its preservation under the corrected pointwise error estimate is explicit in Section 3 of [2]. Write $R_\epsilon = f_\epsilon * f_\epsilon$. The Fourier identity

$$R_\epsilon - \widehat{R_\epsilon''}/(4L^2)(z) = (1 + \pi^2 z^2/L^2)K_\epsilon(z)^2$$

cancels that weight at $z = z_\rho - z_{\rho'}$. Both R_ϵ and R_ϵ'' are fixed real even integrable functions, supported strictly inside $[-1, 1]$ and Lipschitz at zero. Applying the lemma separately to them, then multiplying the latter asymptotic by $1/(4L^2)$, avoids an error estimate for a varying test function. The result is

$$E_\epsilon = (C(f_\epsilon) + o_\epsilon(1))N, \quad C(f_\epsilon) \longrightarrow 2 - H_f. \quad (12)$$

This analytic theorem is an attributed dependency, not a conclusion of the finite interval computations.

To use (10) for the smoothed matrix, write $d_\epsilon = \|f_\epsilon - f\|_1$. Every real kernel entry changes by at most d_ϵ , uniformly in its argument. An m -point block changes by at most md_ϵ in operator norm. Since Ψ_τ is $2 \max(1, \tau)$ -Lipschitz on the nonnegative spectrum, its trace changes by at most $2 \max(1, \tau)m^2 d_\epsilon$. Summing fixed-size blocks costs $O_{m, \tau}(d_\epsilon)N$.

Riemann-von Mangoldt bounds the real low-point span by $TL/(2\pi) = N + o(N)$. Substituting (10) into (11) and comparing with (12) gives

$$(2 - a)N_d \geq (1 + H_f - \beta - O_{m, \tau}(d_\epsilon))N \\ + (6c - 7 - c^2 - a)h + (4c - 2 - c^2 - 2a)k - o_\epsilon(N) - O_m(1).$$

Fix the profile, m, τ, c first, let $T \rightarrow \infty$ next, and only then let $\epsilon \rightarrow 0$. The nonnegative residuals in (6) prove Theorem 2. No growing-block limit or effective height is used.

3 Windows and exact parameter choices

Both profiles have the form

$$v(s) = \cos(\sqrt{2}s) + \sum_{j=1}^J c_j \cos(2\pi j s), \quad f(s) = v(s)/I_1, \quad I_1 = \sqrt{2} \sin(1/\sqrt{2}), \quad (13)$$

on $[-1/2, 1/2]$, and zero outside. The numerators of c_j , all over 10^9 , are listed in Table 1. The nine-point profile is from [5]; the thirteen-term profile is typh's optimized window used by Lavery [4]. Positivity follows from $v(s) \geq \cos(1/\sqrt{2}) - \sum |c_j| > 0$.

j	nine-point coefficient numerator	thirteen-term numerator
1	3322500	12497829
2	-7609135	-14424455
3	1190194	2621498
4	-731476	7207519
5	-1680572	-2561956
6	1141360	6199418
7	0	-7197280
8	0	4324511
9	0	-5325091
10	0	-309374
11	0	443802
12	0	-310460

Table 1: Cosine coefficients in (13); each entry is divided by 10^9 . The leading $\cos(\sqrt{2}s)$ coefficient is one.

Proposition 5 (Scalar identity, D). *With $\theta = 1/\sqrt{2}$ and (13),*

$$H_f = \frac{3}{2} - \frac{\cos \theta}{\sqrt{2} \sin \theta} - \frac{\sum_{j=1}^J c_j^2 (1/2 - 1/(4\pi^2 j^2))}{2 \sin^2 \theta}. \quad (14)$$

Proof. Let $\mathcal{J}$ be the integral operator with kernel $|s - t|$ on $[-1/2, 1/2]$. Solving $(\mathcal{J} \cos(\omega s))'' = 2 \cos(\omega s)$ with its endpoint conditions gives

$$\mathcal{J} \cos(\omega s) = \frac{\sin(\omega/2)}{\omega} + \frac{2 \cos(\omega/2)}{\omega^2} - \frac{2 \cos(\omega s)}{\omega^2}.$$

Consequently $(I + \mathcal{J}) \cos(\sqrt{2}s)$ is constant. Each $\cos(2\pi j s)$ has zero integral, so the cross terms against the ground profile vanish. Orthogonality removes the other cross terms and gives $\langle e_j, (I + \mathcal{J})e_j \rangle = 1/2 - 1/(4\pi^2 j^2)$. Thus

$$\langle v, (I + \mathcal{J})v \rangle = \sin^2 \theta + \sqrt{2} \sin \theta \cos \theta + \sum c_j^2 (1/2 - 1/(4\pi^2 j^2)).$$

Divide by $I_1^2 = 2 \sin^2 \theta$ and subtract from two. □

For the thirteen-term profile an exact rational outward enclosure gives

$$\begin{aligned} 0.6721710926412995494027803172404056748101 &\leq H_f, \\ H_f &\leq 0.6721710926412995494027803172404056748102. \end{aligned}$$

The calculation brackets $\sqrt{2}$ with integer square-root inequalities, π with Machin's formula and forty alternating-series terms, and sine and cosine with twenty-four Taylor terms and explicit remainders. It uses only rational endpoints. A separate Arb calculation evaluates the scalar

	seven-point attributed input	nine-point completed computation
r	6	8
δ	39369/5000000	3051/500000
B	199193/50000000	8/2500
H used	33608554629/50000000000	3362285207/5000000000
m	742	958
τ	12043/5000	2409/1000
c	17043/5000	3409/1000

Table 2: Exact inputs for Theorem 2.

integrals using an ODE-derived formula and native sinc, checking all 169 symmetry comparisons. For the nine-point profile a source-bound Arb calculation with a full 4096-cell profile comparison gives $H_f \geq 3362285207/5000000000$.

For the seven-point input, the six pressure numerators over 10^8 are

$$(41468, 70344, 87381, 87381, 70344, 41468).$$

Lavery’s named theorem `cert_AM` proves (5) on the full nonnegative orthant; `Fw_W7` identifies its functional with the certified expression. The source completes its half-space certificate by reflection. Its metadata attributes the optimized window and evaluator to `typh` and the weighted-point refinement to `Ainta`. The reported external Lean/nanoda checks are preserved, but were not rebuilt locally.

For the seven-point choice,

$$a = \frac{905487}{115937500}, \quad \beta = \frac{4581439}{1159375000}, \quad \tau^2 - D = \frac{155929}{25000000},$$

and the two multiplicity residuals are $16929063061/9275000000$ and $2094101/9275000000$, respectively. The threshold residuals are 0 and $2043/10000$. For the nine-point choice $D = 57969/10000 < 5803281/1000000 = \tau^2$; all remaining rational residuals are likewise nonnegative. Direct fraction arithmetic in (7) proves both constants in Theorem 1.

4 The complete nine-point local certificate

Proposition 6 (Machine-verified local inequality, MV). *For the nine-point window in Table 1 and the exact pair weights in Appendix A, (5) holds for every eight-gap vector $g \geq 0$, with $b_i = 1/2500$ and $\delta = 3051/500000$.*

The computation uses a grid of spacing $1/4000$ and 128-bit Arb enclosures. Beyond the exact pressure cutoff, the nonnegative pressure alone proves the inequality. One-body lower bounds remove whole closed coordinate cells whose pressure plus adjacent overlap already exceeds the target. The remaining Cartesian domain has 256 initial boxes, distributed deterministically across 96 shards. Each node either proves a valid lower bound or splits an integer cell range into disjoint children whose closed real boxes cover the parent. No unresolved terminal cell is accepted.

Three bounds certify a node: the nonnegative pressure bound, a sum of closed-cell kernel lower bounds, or a convex lower bound with a positive Hessian certificate. For the last bound, each pair term contributes a signed lower coefficient times a rank-one positive matrix. Their sum L satisfies $\nabla^2 F(x) \succeq L$ throughout the box. A floating-point factorization selects candidate boxes only; positive definiteness is proved by an Arb LDL factorization of the exact dyadic coefficients.

For a midpoint x_0 and gradient $g = \nabla F(x_0)$, integrating the Hessian on each segment gives

$$F(x) \geq F(x_0) + g^T(x - x_0) + \frac{1}{2}(x - x_0)^T L(x - x_0) \geq F(x_0) - \frac{1}{2}g^T L^{-1}g.$$

If $L = \mathcal{L}\mathcal{D}\mathcal{L}^T$, solve $\mathcal{L}z = g$ and compute $g^T L^{-1}g = \sum z_i^2/\mathcal{D}_{ii}$ by enclosed arithmetic. The unconstrained minimum is a lower bound even if it lies outside the box. The algorithm also retains the first-order box bound and accepts either certified bound. Singular or uncertain pivots reject this path.

All 96 shards complete, with 107,752,902 nodes, 53,876,323 splits and 53,876,579 prunes. The initial-box identity $107752902 = 2(53876323) + 256$ agrees with the binary-tree accounting. The prune types are 95,807 pressure, 30,730,268 interval and 23,050,504 convex-bound decisions. The independent standard-library checker reconstructs the domain, shard allocation, source/table bindings and completed tree counts. Twelve corrupted copies of actual output are rejected, followed by a fresh check of unchanged originals. Dense rational quadratic controls separately compare the LDL solve with exact inversion and reject indefinite, singular and uncertain-pivot inputs.

4.1 Independent whole-cell table comparison

Both stored tables, for w and w'' , have 61,029 cells. Their lower coefficients are independently checked on every closed cell at 256-bit precision through native hypergeometric midpoint jets and Fourier Taylor remainders. This evaluates a different kernel representation but shares FLINT/Arb; it is not an independent interval-arithmetic library.

The identity $\text{sinc}(z) = {}_0F_1(3/2; -z^2/4)$ supplies the midpoint jets. Write the unnormalized transform as a finite combination of shifted sinc functions. If B_0 is the sum of the absolute profile coefficients including the leading one, then $|K''_{\text{raw}}| \leq \pi^2 B_0$. Enclose the raw kernel on a radius- R subcell by its midpoint value, first derivative and this quadratic remainder, then divide by its positive normalization before squaring. For $A_0 = B_0/I_1$, the global bound $|w^{(4)}| \leq (2\pi)^4 A_0^2$ follows by differentiating the Fourier representation of K^2 supported on $[-1, 1]$. Thus a lower enclosure for w'' is

$$w''(x_0) - |w'''(x_0)|R - \frac{1}{2}(2\pi)^4 A_0^2 R^2.$$

Adaptive bisection proves the comparison on all closed subcells, including endpoints. The full comparison completes in 225.276 seconds, using 158,883 subcell enclosures and maximum bisection depth three. Deliberately increased table coefficients are rejected.

5 Reproducibility and limits

The exact-byte archive for computation EXP-020 contains the completed tables, reports, checkpoints, source bindings, actual physical source bytes, the baseline cache and both standard-library checkers. The 233-member ZIP has 1,854,946 bytes and SHA256

91be4798774837bc16007a8236a692078e8
25ffb1c8dfa476c05d3c733b17879.

CRC, member and byte comparisons pass. After extraction, both the complete cover checker and exact transfer checker reproduce the same result. These transport checks do not rerun all interval operations.

The interval execution trusts the frozen Python verifier, python-flint 0.9.0, FLINT/Arb and directed binary64 enclosures. The complete whole-cell comparison strengthens the input-table evidence; the metadata checker proves coverage and binding consistency, not each numerical pruning operation. There is no tangent-free full replay or end-to-end Lean proof for the nine-point computation. Windows full-worker replay is documented; other platforms are not claimed to reproduce its path bindings unchanged.

The seven-point consequence in (2) trusts Lavery’s external local theorem. Its source SHA256 is

65564079527487fde93b43bb83dd840a
768acf9fb8db1f1f015d38c4faf3b7d4.

The Apache-2.0 source and its external verification log and attestation are preserved. An independent pure-rational implementation rechecks the window, source functional and exact transfer. It also checks 288 list/block examples and 183 pair-count examples with the actual unequal pressures; these finite examples supplement the general proof rather than establishing universality.

The computation identifiers refer to reproducible source and outputs under **problems/number-theory/riemann-hypothesis/experiments/** in [the source repository](#). EXP-020 supplies the locally replayed certificate; EXP-025 supplies the vector-pressure implication and independent rational window calculation. Supporting EXP-022 bounds the output of the fixed nine-point window/weight transfer family below $0.8374212877970697\dots$. That ceiling restricts the method with those fixed inputs, not the true zero proportion, and does not restrict other windows or weights.

The counting conclusions are asymptotic and give no effective finite height. They do not prove RH, universal simplicity, a simple-critical-line proportion, or an improved short-interval onset. Reported external formal checks and internal recomputation do not establish peer review. Improvements from incomplete local covers are not used.

A Exact pair weights

The upper-triangular entries below are numerators; unspecified diagonal and lower-triangular entries are zero. Each index span has total weight at most two. For the nine-point table each span mass equals exactly two.

$i \setminus j$	0	1	2	3	4	5	6	7	8
0	–	1802576	4832031	5411933	10000000	10000000	10000000	10000000	20000000
1	–	–	2694869	2295599	1780844	0	0	0	10000000
2	–	–	–	2714860	0	2807223	0	0	10000000
3	–	–	–	–	2787695	5744740	2807223	0	10000000
4	–	–	–	–	–	2787695	0	1780844	10000000
5	–	–	–	–	–	–	2714860	2295599	5411933
6	–	–	–	–	–	–	–	2694869	4832031
7	–	–	–	–	–	–	–	–	1802576

9-point weights: divide each numerator by 10000000.

$i \setminus j$	0	1	2	3	4	5	6
0	–	21269442	890898	28398464	75539688	99999999	200000000
1	–	–	35896622	65972058	71601535	48920622	99999999
2	–	–	–	42833935	66274085	71601535	75539688
3	–	–	–	–	42833935	65972058	28398464
4	–	–	–	–	–	35896622	890898
5	–	–	–	–	–	–	21269442

7-point weights: divide each numerator by 100000000.

References

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